HW0302

■ No. ■ Date

C10501 Q13:

The points on the normal probability plot fall approximately along a straight line

⇒ The data came from a normal population

C10502 Q20:

$H_0: \mu = 74 \quad H_a: \mu \neq 74$

$\mu = 74,\ n = 25,\ df = 24$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{69.7 - 74}{16/\sqrt{25}} = -1.344$$

$df = n - 1 = 24$

$t_{\frac{\alpha}{2},(n-1)} = t_{0.025,24} = 2.064$

$|-1.344| < |2.064| \Rightarrow$ non-reject H_0

C10502 Q30:

a. $H_0: \mu = 25,\ H_a: \mu < 25$

$$t = \frac{20.3 - 25}{\frac{5}{\sqrt{21}}} \doteq -4.31$$

$df = n - 1 = 20,\ t_{0.05,20} \doteq -1.725$

$-4.31 < -1.725 \Rightarrow$ reject H_0

b. $\bar{x} \pm t_{0.025,df} \cdot \frac{s}{\sqrt{n}}$

$= 26.6 \pm t_{0.05,20} \cdot \frac{7.4}{\sqrt{21}}$

$= 26.6 \pm 2.086 \cdot 1.615$

$= 26.6 \pm 3.37 \Rightarrow (23.23,\ 29.97)$

c. The posttreatment and follow-up are higher than pretreatment.

⇒ Self-esteem ↑ after treatment

⇒ treatment is effective

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C10502 Q35

a. $H_0: \mu = 22$, $H_a: \mu > 22$

$$t = \frac{31-22}{\frac{6.2}{\sqrt{17}}} \doteq \frac{9}{1.503} \doteq 5.99$$

$df = n-1 = 16$

$t_{0.05,16} \doteq 1.746$

$5.99 > 1.746 \Rightarrow$ reject H_0

b. C.I.: $31 \pm 1.746 \cdot \frac{6.2}{\sqrt{17}} \Rightarrow (28.37, 33.63)$

C10503 Q7

$H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$

$\bar{x}_1 = 7$, $\bar{x}_2 = 8.6$, $s_1^2 = 15.33$, $s_2^2 = 10.3$, $n_1 = 4$, $n_2 = 5$

$df = n_1 + n_2 - 2 = 7$

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{3 \times 15.33 + 4 \times 10.3}{7} = 12.4571$$

$$t^* = \frac{7-8.6}{\sqrt{12.4571 \cdot (\frac{1}{4}+\frac{1}{5})}} \doteq -0.6958$$

$|t| > t_{0.025,7} = 2.365$

p-value $\doteq 0.5209 > 0.05$ 且 $|t^*| < 2.365$, non-reject H_0

C10503 Q8

$H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$ $df = 16+13-2 = 27$

$\bar{x}_1 = 34.6$, $\bar{x}_2 = 32.2$, $n_1 = 16$, $n_2 = 13$, $s_1^2 = 4.8$, $s_2^2 = 5.9$

$$s^2 = \frac{15 \times 4.8 + 12 \times 5.9}{27} \doteq 5.2889$$

$$t^* = \frac{34.6 - 32.2}{\sqrt{5.2889(\frac{1}{16}+\frac{1}{13})}} \doteq 2.795$$

p-value $\doteq 0.00944$

$|t| > t_{0.025,27} \doteq 2.052$

$|t^*| > 2.052$ 且 $p < 0.05 \rightarrow$ reject H_0

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C10503 Q12

a. $H_0: \mu_1 = \mu_2$, $H_a: \mu_1 \neq \mu_2$

$$s = \sqrt{\frac{15^2}{10} + \frac{12^2}{10}} \doteqdot 6.07$$

$$t^* = \frac{86.4 - 74.7}{6.07} \doteqdot 1.10$$

$df = 10 + 10 - 2 = 18 \Rightarrow t_{0.975, 18} \doteqdot 2.101$

$1.10 < 2.101 \Rightarrow$ non-reject H_0

b. $(\bar{x}_1 - \bar{x}_2) \pm t_{0.975, df} \cdot SE$

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{8.95^2}{10} + \frac{10.9^2}{10}} \doteqdot 4.411$$

$\Rightarrow 6.9 \pm 2.101 \times 4.411$

$\Rightarrow (-2.56, 15.96)$

c. 區間包含 0 ⇒ 差異不顯著

evidence 不足以證明 iPad 有用，

可能只是隨機誤差